

State-of-the-Art Analysis of Attention-Based and Structured Filter Pruning in Convolutional Neural Networks

Research context:

Feedback-Driven Filter Pruning for Deep Convolutional Networks

Publications from 2024–2026

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Course: Computational Intelligence

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1 Introduction

Deep convolutional neural networks (CNNs) achieve high predictive accuracy at the cost of substantial parameter storage, arithmetic complexity, memory traffic, and inference latency. These costs restrict their use on mobile and edge devices. Neural-network compression addresses this problem through pruning, quantization, low-rank decomposition, and knowledge distillation. The research topic motivating this report focuses primarily on *structured filter pruning*: removing complete convolutional filters and their corresponding output channels so that the result is a smaller dense network that can run on standard hardware without dedicated sparse kernels.

The central difficulty is not the physical removal of a filter, but deciding which filters can be removed without eliminating information required by the task. Classical criteria based only on weight magnitude or local activation statistics are inexpensive, but they may confuse a weak numerical response with low semantic importance. This is particularly problematic in early CNN layers: an apparently unimportant low-level feature can become useful after it is combined with features in deeper layers. A task-aware selection mechanism should therefore consider not only the current layer but also the effect of its channels on later representations and the final prediction.

1.1 Attention as a Compression Mechanism

In this context, *attention* does not necessarily mean Transformer self-attention. It denotes a learned importance mechanism that assigns a score to each convolutional channel. Let $A^{(l)} \in R^{C_l \times H_l \times W_l}$ be the activation tensor of layer l . A channel descriptor can be obtained by global pooling,

$$z_c^{(l)} = \frac{1}{H_l W_l} \sum_{i=1}^{H_l} \sum_{j=1}^{W_l} A_{cij}^{(l)}, \quad (1)$$

and a lightweight gating network can convert the descriptors into scores $s^{(l)} \in [0, 1]^{C_l}$. The activations are then reweighted as

$$\tilde{A}_c^{(l)} = s_c^{(l)} A_c^{(l)}. \quad (2)$$

Sparsity regularization, a target resource constraint, or a discrete selection operator encourages some scores to approach zero. After training, a hard mask can remove the associated filters and the corresponding input channels of the next layer. Attention therefore serves as a trainable pruning criterion rather than merely as a feature-enhancement block. Its value is that filter relevance can be optimized using the task loss instead of being defined by a fixed handcrafted norm.

Three forms of channel selection are especially relevant:

- **static learned selection** produces one set of channel scores for all inputs and is well suited to permanent structural pruning;
- **input-dependent selection** computes different scores for different samples, increasing flexibility but usually retaining the full model in memory; and
- **feedback-driven selection** additionally uses information from deeper layers or the final task response to revise the importance of earlier filters.

The third form is the proposed research direction. To distinguish this planned idea from the methods reviewed later, the following expression is only a conceptual formulation of a possible feedback-aware gate, not an equation taken from an existing publication:

$$s^{(l)} = g_{\theta_l} \left(z^{(l)}, f^{(l+1:L)} \right), \quad (3)$$

where $f^{(l+1:L)}$ summarizes information from deeper layers. The feedback signal supplies semantic context that a purely local score cannot observe. A practical method must obtain this benefit without introducing an attention module whose cost cancels the savings produced by pruning.

1.2 Research Motivation and Objectives

The planned research investigates whether learned channel attention and deeper-to-shallower feedback can improve filter selection in residual CNNs. It compares static masks, input-dependent attention, feedback-based cross-layer gates, and differentiable discrete selection. A further objective is to learn the number of retained filters per layer instead of imposing a uniform pruning ratio manually. The main evaluation backbones are ResNet-18/34/50, with CIFAR-10/100, ImageNet-100, and ImageNet-1K providing increasingly demanding classification settings.

The primary outcome is a permanently smaller structured model. Accuracy must therefore be reported together with parameter count, floating-point operations (FLOPs), multiply-accumulate operations (MACs), measured latency, model size, and the layer-wise pruning pattern. After filter pruning and fine-tuning, the resulting network may also be combined with low-rank factorization and 8-bit integer (INT8) quantization. These later stages complement attention: attention decides *which structures are necessary*, whereas low-rank and low-precision methods reduce the cost of the structures that remain.

2 Review Scope and Comparison Protocol

The state-of-the-art (SOTA) table contains 12 peer-reviewed publications from 2024–2026. Its core comprises methods that learn filter importance, perform input-dependent channel selection, or account for cross-layer channel dependencies. Broader structured-pruning methods are retained as contextual baselines for architecture search, resource allocation, and deployable acceleration; they are not presented as feedback-based methods. Transformer key–value-cache compression and purely unstructured weight sparsity are outside the main scope because they do not directly address the problem of removing CNN filters.

Publication metadata and numerical results are taken from the original papers or their official proceedings pages. The final publication year is used. The table reproduces representative author-reported results; it should not be interpreted as a single ranking because the papers use different networks, datasets, hardware, training recipes, and compression budgets. Reductions are relative to the corresponding dense baseline unless stated otherwise.

3 State-of-the-Art Comparison

The table below summarizes the architectures, contributions, and representative results of the 12 selected publications.

Table 1: Recent methods relevant to attention-based and structured filter pruning in CNNs.

Year	First author	Model / method	Technology	Main contribution	Representative metrics / score
2026	Razi Iqbal	Attention-Driven Structured Compression (ADSC) [1]	Four-layer CNN; global activation statistics; two-layer multilayer perceptron with sigmoid filter scores; teacher–student weight transfer.	Introduces a lightweight, task-aware attention scorer trained from one mini-batch. The scorer replaces fixed magnitude criteria and selects filters before constructing a compact student, with diversity regularization to avoid retaining redundant channels.	The authors report 56.71% CIFAR-100 accuracy versus 56.03% for the teacher (+0.68 pp), with 50.5% parameter compression. On CIFAR-10, they report 85.79% accuracy at 51.0% compression.
2026	Huiling Jiang	Connection-Aware DenseNet Pruning (CADP) [2]	DenseNet-121; EdgePrune using activation correlation and gradients; connection-guided canonical polyadic tensor decomposition; dual-stream knowledge distillation with spatial-attention transfer.	Treats inter-layer feature paths as the pruning unit and uses their importance to allocate layer-specific tensor ranks. It integrates structured pruning, low-rank decomposition, and knowledge recovery in one pipeline.	The authors report 88% parameter reduction, 89% computational savings, 99.67% accuracy on PlantVillage, and 99.66% on RiceLeaf. The evidence is limited to DenseNet-121 and two controlled plant-disease datasets; results are means from five-fold cross-validation.
2025	Anouar Chahbouni	Dynamic Attention-Guided Pruning (DAGP) [3]	VGG-16, ResNet-56, and a custom CNN; trainable sigmoid masks after convolution; delayed ℓ_1 sparsity; optional hard channel removal.	Embeds differentiable masks directly in the CNN and jointly optimizes classification and sparsity. The masks act as implicit attention and remove the need for a separate prune–retrain cycle.	For the custom CNN on PlantVillage, the authors report +1.35 pp accuracy, 99.3% parameter reduction, and 98.82% FLOPs reduction. Their Raspberry Pi 4 validation reports 60 ms inference and 2.00 MB RAM.
2025	Peyman Rostami	Information-Theoretic Pruning of Coupled Channels (ITPCC) [4]	ResNets; variational information bottleneck; probabilistic importance of structurally coupled channel sets.	Models channels that must be removed together because of residual and skip connections. It prunes a coupled group only when the complete group carries little task-relevant information, avoiding invalid tensor configurations.	For CIFAR-100/ResNet-56, the illustrated configuration is reported to reach 71.06% accuracy with 66.75% FLOPs reduction; the comparison method reaches 65.33% and produces an incompatible structure.

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Year	First author	Model / method	Technology	Main contribution	Representative metrics / score
2025	Jiamu Zhang	Flexible Group Count / LeanFlex Grouped-Kernel Pruning (LeanFlex-GKP) [5]	VGG and ResNet; grouped convolution; flexible Conv2d (groups) count; integral optimization; one-shot, post-training, data-agnostic pruning.	Makes group count the adaptive variable in grouped-kernel pruning. The output remains a standard dense grouped-convolution model and requires little manual criterion or hyperparameter design.	For CIFAR-100/ResNet-56, the authors report 72.11% Top-1 (+0.58 pp), with 43.22% MAC reduction and 43.18% parameter reduction.
2025	Xinglong Sun	Multi-Dimensional Pruning (MDP) [6]	CNNs and vision Transformers; latency lookup tables; saliency estimation; mixed-integer nonlinear programming across channels, heads, embeddings, and blocks.	Optimizes several structural dimensions jointly under a measured latency constraint rather than treating FLOPs as a proxy. It automatically searches for a hardware-aware pruned configuration.	For ImageNet/ResNet-50, the authors report a 28% speed increase and +1.4 pp Top-1 relative to Hardware-Aware Latency Pruning (HALP). On CPU, their MDP model reaches 118.2 FPS and 75.2% Top-1, versus 45.9 FPS and 74.5% for HALP.
2024	Shangqian Gao	BilevelPruning: Unified Dynamic and Static Pruning (UDSP) [7]	ResNet and MobileNetV2; differentiable channel gates; bilevel optimization; combined static and input-dependent dynamic pruning.	Uses dynamic subnetworks to optimize a permanently reduced static subnetwork. The method therefore reduces stored parameters while retaining per-input channel-selection flexibility during inference.	For ImageNet/ResNet-50, the authors report 76.51% Top-1 (+0.38 pp), with 58.4% FLOPs reduction and 25.0% parameter reduction.
2024	Ashish Kumar	Pick-or-Mix (PiX) [8]	ResNet and other ConvNets; channel grouping; activation-conditioned, pixel-wise pick-or-mix sampling; replacement of 1×1 convolutions.	Selects or mixes one output from each channel subset according to the current feature map. It supplies input-dependent channel reduction using ordinary dense operations rather than custom sparse kernels.	The authors report that replacing ResNet 1×1 channel-squeezing layers makes the network 25% faster without an accuracy loss; those layers account for about 25% of ResNet-50 FLOPs in their analysis.

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Year	First author	Model / method	Technology	Main contribution	Representative metrics / score
2024	Rinor Cakaj	CNN Mixture-of-Depths (MoD) [9]	ResNet; learned channel router; selected channels traverse a convolutional block while the remainder bypass it; fixed tensor sizes and static graph.	Introduces conditional channel computation without custom GPU kernels, special losses, or post-training fine-tuning. It converts dynamic selection into a hardware-friendly implementation.	For ImageNet, the authors report 76.27% Top-1 for ResNet75-MoD versus 76.25% for ResNet-50, 3.48 versus 4.13 billion MACs, and speedups of $1.25\times$ on CPU and $1.15\times$ on GPU.
2024	Alireza Ganj-danesh	Reinforcement Learning and Soft Alignment (RLAL) [10]	ResNet and MobileNetV2; reinforcement-learning agent for layer pruning ratios; epoch embeddings, gated recurrent unit environment model, and weight-subnetwork alignment.	Jointly trains CNN weights and discovers the structured architecture, avoiding the need for a pretrained model. The recurrent component represents the non-stationary reward produced while the CNN itself is learning.	For CIFAR-10/ResNet-56, the authors report 93.86% Top-1 (+0.45 pp) with 50.0% FLOPs reduction. For MobileNetV2, they report 94.85% (+0.37 pp) with 49.4% FLOPs reduction.
2024	Donghyeon Lee	Nuclear-Norm Shared Pruning Module (NuSPM) [11]	ResNet; shared pruning modules for residual-channel alignment; nuclear-norm and MAC-aware regularization; training from scratch.	Solves channel-alignment constraints in residual blocks and preserves representational capacity during pruning. The structure and weights are learned together without a pretrained network.	For ImageNet/ResNet-50, the authors report 75.25% Top-1 while retaining 41% of the original MACs, equivalent to a 59% MAC reduction.
2024	Parakh Agarwal	Soft Pruning with Weight Blending (SPWB) [12]	ResNet and MobileNetV2; soft pruning by weight blending; connection-aware channel removal; standard dense deployment.	Moves selected weights gradually toward zero instead of deleting channels abruptly, giving the network time to adapt and accounting for dependencies between adjacent layers.	For ImageNet/ResNet-50, the authors report approximately 50% FLOPs reduction with a 0.65 pp Top-1 accuracy decrease.

4 Synthesis and Research Gap

The recent literature shows a clear shift from fixed local heuristics toward learned, task-aware importance estimation. DAGP and ADSC are the closest direct examples of attention used as a pruning criterion: both learn filter scores, but DAGP primarily learns global mask parameters, whereas ADSC derives a static student from activation statistics. PiX, CNN MoD, and BilevelPruning demonstrate the benefit of input-dependent channel decisions, although dynamic execution normally preserves more weights in memory than a permanently pruned model.

Not every method in Table 1 is attention-based. LeanFlex-GKP, NuSPM, SPWB, and MDP are contextual structured-pruning baselines: they clarify how channel grouping, residual constraints, gradual removal, and measured latency affect deployability. Their role is to define comparison criteria for the planned method, not to provide evidence for feedback-driven selection.

The results also show that channels cannot always be evaluated independently. ITPCC enforces the structural coupling created by residual connections, while CADP explicitly evaluates inter-layer paths and transfers intermediate spatial attention during distillation. These methods support the premise that local filter magnitude is an incomplete importance signal. CADP nevertheless reports results only for DenseNet-121 on two controlled plant-disease datasets, so it supports the architectural motivation rather than establishing general performance on residual networks or ImageNet. Moreover, connection-aware pruning is not the same as closed-loop feedback: the reviewed methods do not systematically use a deeper semantic representation to update an earlier layer’s input-dependent filter decision and then convert those decisions into an automatically sized dense CNN.

This leaves a focused research opportunity at the intersection of the three method families. A feedback-driven pruning method should combine:

- task-aware, input-dependent channel attention;
- explicit information flow from deeper to shallower pruning gates;
- differentiable or discrete learning of the retained filter count in each layer;
- hard structural removal for real reductions in parameters and latency; and
- compatibility with later low-rank factorization and quantization.

Finally, the table confirms that FLOPs alone are not a sufficient deployment metric. PiX and CNN MoD emphasize implementation-friendly execution, MDP optimizes measured latency directly, and DAGP includes an edge-device test. Consequently, the planned evaluation should distinguish theoretical compression from realized speedup and report accuracy, parameters, FLOPs, latency, memory, and compressed file size under the same experimental setup.

5 Conclusions

Attention is useful in CNN compression when it is treated as a learned estimate of filter importance. Static attention supports permanent channel removal; input-dependent attention improves flexibility; and feedback from deeper layers may supply the semantic context missing from local pruning criteria. The 12 publications reviewed here establish recent baselines for these components, but they do not yet combine all of them in a single deployable framework. This gap directly motivates feedback-driven filter pruning and provides a clear basis for comparing the proposed method with current SOTA approaches from 2024–2026.

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